

BMJ Open Comparative efficacy and safety of different surgical approaches for the treatment of anterior cruciate ligament injury: a Bayesian network meta-analysis protocol

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ABSTRACT

Introduction Anterior cruciate ligament (ACL) rupture can be treated surgically or non-surgically, with several surgical interventions available at present. However, the comparatively effective surgical intervention with relatively fewer side effects remains unknown. This study aims to fill in this gap by conducting a Bayesian network meta-analysis (NMA) and provide a theoretical basis for the clinical application.

Method and analysis We will perform a Bayesian NMA and will include randomised controlled trials (RCTs) published in English or Chinese that compare surgical intervention (ie, standard ACL reconstruction, ACL remnant-preserving reconstruction and ACL repair with suture augmentation to conservative therapy or studies that compare one surgical intervention to another for the symptom relief and function recovery of patients with ACL rupture. Primary outcome will be the proportion of patients with symptomatic and functional improvement measured by the Knee Injury and Osteoarthritis Outcome Score before and 6 months after treatment, with scores ranging from 0 (worst) to 100 (best). Secondary outcomes will be knee-specific quality of life (ACL QoL), return to activity and level of sport participation (Tegner or modified Tegner score), health-related QoL (EuroQol Group 5-Dimension 5-Level, EQ-5D-5L), resource use, intervention-related complications and patient satisfaction. We have developed search strategies for PubMed, Embase, the Cochrane Library and Web of Science, retrieving RCTs that meet the inclusion criteria from database inception to 1 December 2023. The methodological quality of the included RCTs will be assessed based on the Cochrane risk of bias table. The relative ranking probability of the best intervention will be estimated using the surface under the cumulative ranking curve. The Bayesian NMA will be conducted by using WinBUGS V.1.4.3. The Grading of Recommendations Assessment, Development and Evaluation approach will be applied to determine our confidence in an overall treatment ranking from the NMA.

Ethics and dissemination Ethical approval for this study is not required because no private or confidential patient data will be used in this study. Findings of this study would be disseminated through the publication in a peer-reviewed medical journal.

PROSPERO registration number CRD42023437115.

STRENGTHS AND LIMITATIONS OF THIS STUDY

- ⇒ Rigorous methods will be followed, and the quality of all included articles will be assessed using validated tools.
- ⇒ Literature published in both English and Chinese will be included which broaden the searching scope and knowledge span.
- ⇒ Only randomised controlled trials will be included for analysis, which could limit the evidence by excluding observational studies.
- ⇒ Due to the long-time span of the included literature, multiple factors may contribute to the heterogeneity of target indicators and affect the results.

INTRODUCTION

Anterior cruciate ligament (ACL) rupture predominantly affects young, athletically active individuals, with over 200 000 cases annually in the USA, incurring costs exceeding US\$7 billion.¹ Risk factors of ACL rupture include female sex (with risk two times higher than male),² young age (with a peak at 16 to 18 years), and earlier, more intense, and more frequent participation in sports.³ ACL injuries are often complicated by concomitant injury of the medial collateral ligament (19%–38%) and lateral (20%–45%) or medial (0%–28%) meniscal tears,⁴ leading to symptoms like pain, activity limitation and knee joint swelling.⁵ Diagnosis is effectively done through MRI, which boasts high accuracy.⁶ The injury significantly impacts knee stability and function, decreasing activity levels and life quality, and increasing the risk of long-term osteoarthritis if not properly treated.^{5 7} Despite its prevalence, treatment varies greatly due to the scarcity of high-quality evidence guiding management strategies.

While surgical intervention is frequently favoured for treating ACL ruptures, owing to

its generally better functional outcomes, the choice of the specific surgical technique remains a subject of debate due to the distinct advantages and drawbacks associated with each method.⁸ Common surgical techniques include standard ACL reconstruction, ACL remnant-preserving reconstruction and ACL repair with suture augmentation are common choices.⁹ Each surgical technique for ACL rupture has its merits in restoring joint stability and function. For example, standard ACL reconstruction proven track record of restoring knee stability and function. It has high success rates in terms of patient satisfaction and return to sports. However, it also faces the challenges of graft failure and longer rehabilitation period. But it is a technically more challenging procedure with the risk of preserving damaged tissue that could impair healing.¹⁰ ACL remnant-preserving reconstruction is believed to enhance proprioceptive recovery, improve vascularisation and healing of the graft and potentially less postoperative pain and faster recovery.¹¹ ACL repair with suture augmentation is minimally invasive compared with full reconstruction. Preservation of original ACL tissue can maintain native anatomy and potentially promote quicker recovery and cause less postoperative pain. But it is suitable mainly for proximal ACL tears with good tissue quality and presents with higher rates of failure reported in some studies.^{12 13} When all three surgical methods can be used to treat the same patient, it often leads to difficulty in decision-making for the doctor. Currently, there is a lack of comparative studies to establish which method is superior in terms of effectiveness and safety, an essential consideration for refining ACL treatment strategies.

To fill the gap in this field and provide optimised interventions to patients with ACL rupture, we intend to conduct a Bayesian network meta-analysis (NMA) to evaluate the comparative efficacy of different surgical approaches for the treatment of ACL rupture and provide important guidance to clinical practice.

METHODS AND ANALYSIS

Literature search

This protocol was registered on the PROSPERO International Prospective Register of Systematic Reviews (registration number: CRD42023437115) and developed in accordance with the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols guidelines.¹⁴ Any modifications to the protocol will be updated and subsequently published alongside the results of the NMA. PubMed database, Web of Science database, the Cochrane Central Register of Controlled Trials (CENTRAL) and EMBASE database will be comprehensively searched from the inception of each database up to 1 December 2023 for relevant randomised controlled trials (RCTs). Additionally, we will manually search the references of retrieved papers and reviews to include further relevant material. Grey literature sources such as GreyNet, SIGLE, BLDSC and registry sites like the ICRT Registry Network will also be explored for pertinent

RCTs. Studies selected for further investigation include those comparing surgical interventions for the treatment of ACL rupture, including standard ACL reconstruction, ACL remnant-preserving reconstruction and ACL repair with suture augmentation. We will include RCTs to assess the efficacy and safety of these surgical approaches. The retrieval of literature and data analysis have been completed. We started this study on 20 May 2023 and anticipate completing the submission process by 31 March 2024. During this time, we will continue to screen and incorporate newly published literature into our analysis. Detailed search strategies are listed in online supplemental appendix 1.

Inclusion and exclusion criteria

RCTs that meet the following criteria will be included in this study: (1) the surgical methods for treating ACL rupture are standard ACL reconstruction, ACL remnant-preserving reconstruction or ACL repair with suture augmentation. (2) Compare surgical interventions to conservative therapies or studies that compare one surgical intervention versus another. (3) Studies with follow-up more than 6 months. (4) Studies published in English or Chinese. Exclusion criteria include (1) Non-RCTs, including case reports, reviews, animal trials, letters to the editor and retrospective studies. (2) RCTs from which data cannot be extracted. (3) RCTs published neither in English nor Chinese. Two independent reviewers will perform the study selection. Qualified RCTs will be identified for inclusion by two independent reviewers based on predefined eligibility criteria. Any disagreements at the selection process will be resolved through discussion, or with a third researcher.

Data extraction

Two independent reviewers will extract pertinent data from the included studies. In cases of incomplete data or when the full text is unavailable, we will try to contact the corresponding author of the potentially relevant study. Data presented solely in visual format within figures will be extracted using GetData V.2.20. Data extracted for analysis will include the name of the first author, year of publication, specific details of the populations, type of surgical interventions, outcomes, follow-up time points and side effects of each study. Any disagreements will be solved through discussion or with a third reviewer.

The risk of bias of included RCTs will be assessed according to Cochrane Handbook for Systematic Reviews of Interventions and bias adjustment methods described by Dias *et al.*¹⁵ Bias caused by random sequence generation, allocation concealment, blinding of participants, blinding of outcome assessment, incomplete outcome data, selective reporting and others (mainly including conflict of interests) will be carefully evaluated. Each risk of bias category will be classified using a three-level rating system: low risk, unclear risk and high risk. Studies identified with three or more high-risk factors will be considered to have poor methodological quality.¹⁶ Following the

analysis, trials with three or more high risk of bias will be excluded to determine if the results would change substantially.

Primary outcome will be the proportion of patients with symptomatic and functional improvement measured by the Knee Injury and Osteoarthritis Outcome Score (KOOS) before and 6 months after treatment, with scores ranging from 0 (worst) to 100 (best). KOOS is a wide-accepted patient-reported outcome measure designed to assess the patient's opinion about their knee and associated problems. It consists of five subscales: Pain, Other Symptoms, Function in Daily Living, Function in Sport and Recreation, and Knee-Related Quality of Life (QoL). KOOS has been proven to be reliable and sensitive in assessing outcomes related to knee injuries and osteoarthritis. We will compare the KOOS scores before and 6 months after treatment to assess the improvement or decline in knee health and associated factors. Secondary outcomes will be knee-specific QoL (ACL QoL), return to activity and level of sport participation (Tegner or modified Tegner score), health-related QoL (EuroQol Group 5-Dimension 5-Level, EQ-5D-5L), resource use, intervention-related complications and patient satisfaction. ACL QoL is a questionnaire specifically designed to assess the QoL of patients with ACL injuries. The Tegner Activity Scale is a tool for grading work and sporting activities on a scale from 0 to 10, with higher scores indicating higher activity levels. EQ-5D-5L is a standardised instrument for measuring generic health status. It provides a simple descriptive profile and a single index value for health status. This provides a broader measure of the treatment's impact on overall health and well-being, beyond the specific focus on the knee. Resource use entails the quantification of all healthcare resources used during the treatment and recovery process, including hospital stays, medication and physiotherapy sessions. Understanding the resource implications is vital for assessing the cost-effectiveness and practicality of the treatment. Considering that safety and risk assessment are crucial for a comprehensive evaluation of the treatment's viability and patient advisability, we analysis the intervention-related complications which include any adverse events or complications directly related to the treatment. Patient satisfaction is a subjective measure of the patients' satisfaction with the treatment outcomes and the process. It is an important indicator of treatment success and can influence future treatment choices and adherence.

Statistical analyses

The statistical analyse approaches have been described in detail in our previous paper.¹⁷ Given that Bayesian NMA can narrow the width of the credible interval (CrI) of the estimate by incorporating more studies into each group and shows strength in comparing the efficacy and safety of different surgical approaches,^{18–20} we decided to conduct a Bayesian NMA to identify the most effective and safest intervention.

The posterior density of unknown variables will be estimated using the Markov Chain Monte Carlo method and the variance between RCTs will be accommodated by employing a random effects model.^{20–22} Two Markov chains will be designed to run simultaneously. Each of them starts with a different set of initial values that are selected arbitrarily for convergence. For each set of initial values, a total of 50 000 simulations will be conducted with the first 10 000 simulations being abandoned as the burn-in period. The WinBUGS code can be found at <http://www.bristol.ac.uk/social-community-medicine/projects/mpes/>.

The pooled effect sizes, STD mean difference (SMD) or risk ratio (RRs) will be calculated from the median of the posterior distribution. The 97.5th and 2.5th percentiles of the posterior distribution will represent the upper and lower limit of the 95% CrI, respectively. A 95% CrI including 0 for SMD or 1 for RR will be interpreted as indicating a non-statistically significant difference. In our models, each trial arm will contribute one independent data point, which will help in assessing the model's fit to the data. Posterior mean residual deviance was calculated to measure the fit of the model to the data.²³ If the mean of the residual deviance will be equal or similar to the number of data points of the model, the model fitted the data well.²³ The minimum clinically important difference (MCID) is defined as the smallest change in an outcome that constitutes a clinically meaningful treatment effect,²⁴ which is fundamental to interpreting clinical trial outcomes, making clinical decisions and designing studies with sufficient statistical power to detect any such effect.²⁵ MCID will be defined as 0.5 SD greater change of the related 95% CI.

Risk of bias will be assessed using the Collaboration's 'risk of bias' tool as described above and bias adjustment methods described by Dias *et al* will be applied to evaluate internal validity.¹⁵ We will make use of the model fit and heterogeneity parameters to choose suitable model.²⁶ We will assess the potential publication bias by considering the completeness of the literature search and use funnel plot and regression tests to specify publication bias assessments. Egger's weighted regression and Begg's rank correlation tests will be used to evaluate publication bias. To obtain a meaningful comparison-adjusted funnel plot, we will order the intervention groups by assuming that small studies are likely to favour advanced support surfaces. The certainty of the evidence from the NMA for each network contrast and the ranking of intervention groups was assessed following the GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach.^{27 28} The certainty of evidence from direct comparisons in NMA will be rated as high, moderate, low or very low based on risk of bias, imprecision, inconsistency and publication bias. For indirect comparisons, the quality of evidence will be assessed using GRADE categories and transitivity considerations, with the lower of the two direct comparison ratings applied to the indirect evidence. Confidence in each NMA effect estimate

will be based on the higher rating of direct and indirect evidence but may be downgraded if inconsistencies are found between the two.

Surgical operations will be ranked based on the posterior probability of their effect sizes, with the ranking outcomes presented by the surface under the cumulative ranking curve (SUCRA). A SUCRA value of 100% signifies that the intervention is the most effective treatment, whereas a value of 0% denotes the least effective treatment.^{29 30} We will perform the NMA using WinBUGS (V.1.4.3, MRC Biostatistics Unit, Cambridge, UK). SUCRA figures will be drawn in Stata software (V.15.1, StataCorp).

Patient and public involvement

None.

DISCUSSION

Previous research has indicated that standard ACL reconstruction,³¹ ACL remnant-preserving reconstruction^{32 33} and ACL repair with suture augmentation³⁴ are effective in restoring joint stability of patients with ACL rupture. However, earlier systematic reviews and meta-analyses typically focused on pooling data from trials comparing just two surgical interventions, leaving a gap in comprehensive comparative analysis and an unclear understanding of the relative merits of each method.^{35–37} No previous study has systematically evaluated the symptom alleviation and side effects across all these surgical options. To address this gap, we are conducting this NMA, which allows for an extended comparison across multiple surgical interventions. Bayesian NMA is able to compare the efficacy among different surgical interventions and narrow the width of CrI of the estimate via increasing the number of studies of each group. Our study is pioneering in its approach as the first NMA to compare these surgical interventions and would provide a theoretical basis for clinical application.

ETHICS AND DISSEMINATION

Ethical approval for this study is not required, for there are no private or confidential patient data included in this study. The results would be published in a peer-reviewed medical journal.

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Contributors WZ conceptualised the study, acquired funding and administered the project. YF and YH wrote the original draft. YF and YH reviewed and edited the manuscript.

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Competing interests None declared.

Patient and public involvement Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

Patient consent for publication Not applicable.

Provenance and peer review Not commissioned; externally peer reviewed.

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The Cochrane Library (Wiley Online Library)

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Grey literature: Websites searched

Key words as “anterior cruciate ligament reconstruction” OR “anterior cruciate ligament repair” OR “ACL reconstruction” OR “ACL repair” will be searched in the following grey literature websites.

OpenGrey (www.opengrey.eu/)

GreyNet International (<http://greynet.org/>)

OpenDOAR(<http://www.opendoar.org/>)

WHO (www.who.int)

Bielefeld Academic Search Engine (<http://www.base-search.net/about/en/BASE>)

The National Technical Information Service (<http://www.ntis.gov/>)

NHS UK (www.nhs.uk)

Duke University Medical Center Library (<http://guides.mclibrary.duke.edu/greyliterature>)

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item
ADMINISTRATIVE INFORMATION		
Title:		
Identification	1a	Identify the report as a protocol of a systematic review
Update	1b	If the protocol is for an update of a previous systematic review, identify as such
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number
Authors:		
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments
Support:		
Sources	5a	Indicate sources of financial or other support for the review
Sponsor	5b	Provide name for the review funder and/or sponsor
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol
INTRODUCTION		
Rationale	6	Describe the rationale for the review in the context of what is already known
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)
METHODS		
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated
Study records:		
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review

Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2 , Kendall's τ)
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

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